Symulacja i analiza rozkładów cząstek pp @ $\sqrt{s} = 13.6$ TeV

Pythia 8 + ROOT

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Ustawienia i metoda

Generator	Pythia 8.312, `SoftQCD:inelastic` (minimum bias)
Energia	`Beams:eCM = 13600` GeV, pp
Zdarzenia	100 000, `Random:seed = 12345`
Cząstki	tylko stan końcowy (`isFinal()`)
Kategorie	wszystkie, naładowane, $\pi^{\pm}$ (± 211), $K^{\pm}$ (± 321), $p, \bar{p}$ (± 2212)
1. `generate`	Pythia 8 $\rightarrow$ `events.root` (TTree)
2. `analyze`	`events.root` $\rightarrow$ `analysis.root` (`hN_*, hPt_*, hEta_*, hPhi_*`)
3. `draw_plots.C`	`analysis.root` $\rightarrow$ wykresy PNG i PDF

Pythia podaje $\phi \in (-\pi, \pi]$, więc histogramowane jest $|\phi|$. W `hN_all` 12.5% zdarzeń ma $N > 400$ (overflow).

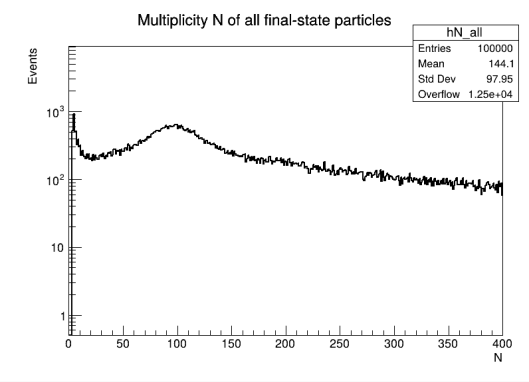
Histogramy: zakresy i biny

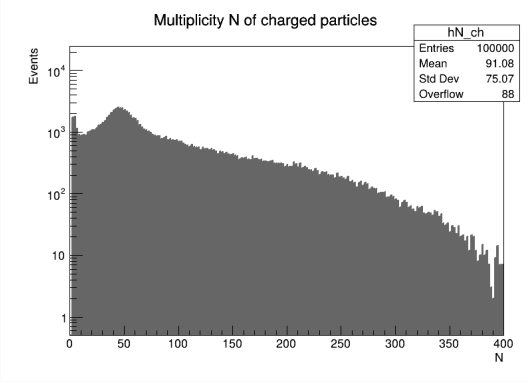
Histogram	Wielkość	Zakres	Liczba binów
`hN_*`	krotność N na zdarzenie	0–400	400
`hPt_*`	pęd poprzeczny p_{T}	0–10 GeV	100
`hEta_*`	pseudorapidity η	-5–5	100
`hPhi_*`	kąt azymutalny $\lvert\phi\rvert$	0–π rad	64

$*$ = `all`, `ch`, `pip`, `pim`, `Kp`, `Km`, `p`, `pbar` (8 kategorii $\times$ 4 wielkości = 32 histogramy)

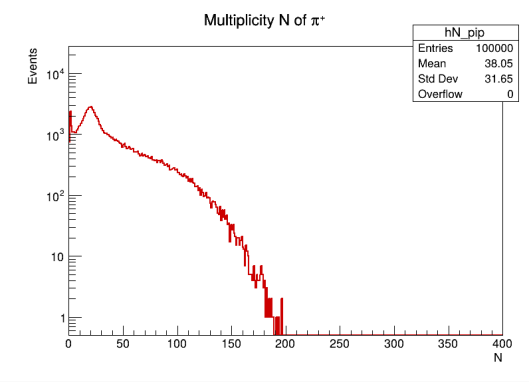
Dodatkowo 4 wykresy porównawcze (`compare_N`, `compare_Pt`, `compare_Eta`, `compare_Phi`): $\pi^{\pm}$, $K^{\pm}$, p i $\bar{p}$ na jednym wykresie.

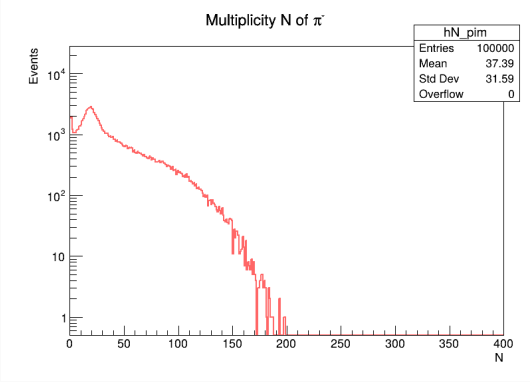
Krotność N: wszystkie i naładowane



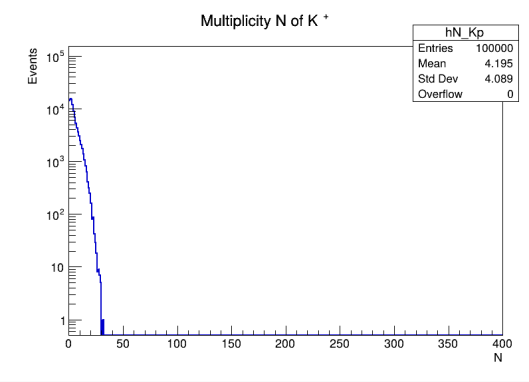


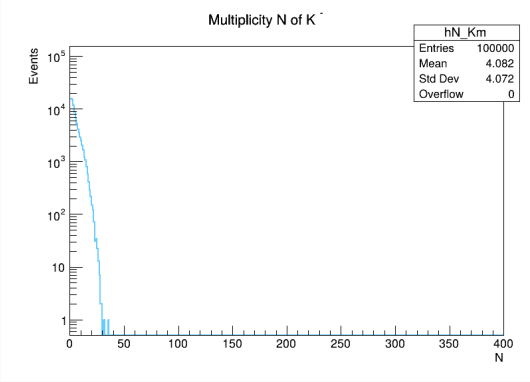
Krotność N: π^+ i π^-



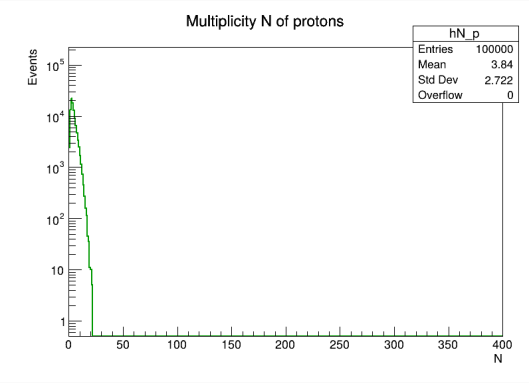


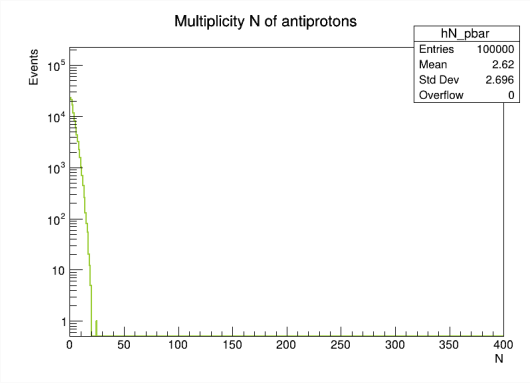
Krotność N: K^+ i K^-



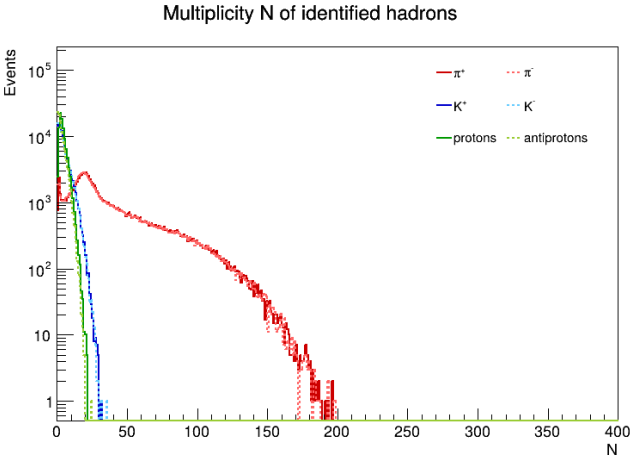


Krotność N: p i $\bar{p}$

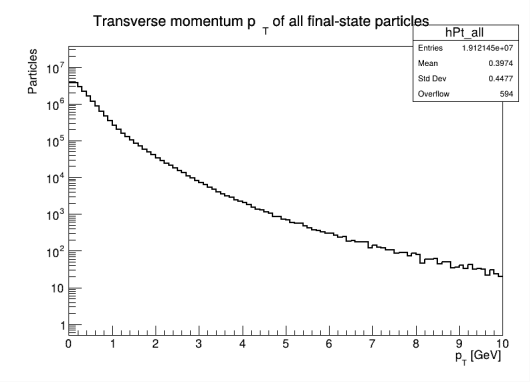


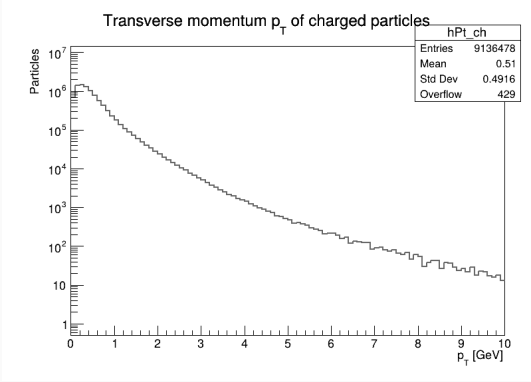


Krotność N: porównanie

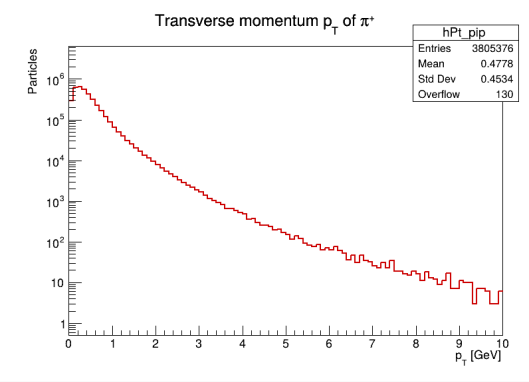


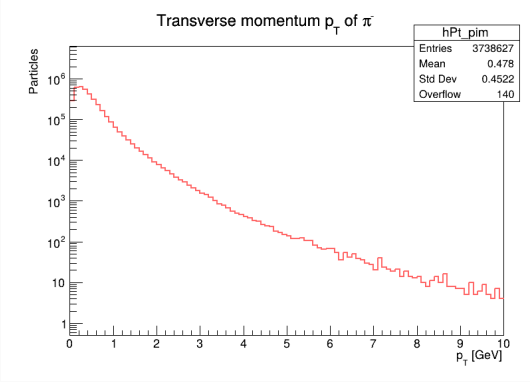
Pęd poprzeczny p_{T}: wszystkie i naładowane



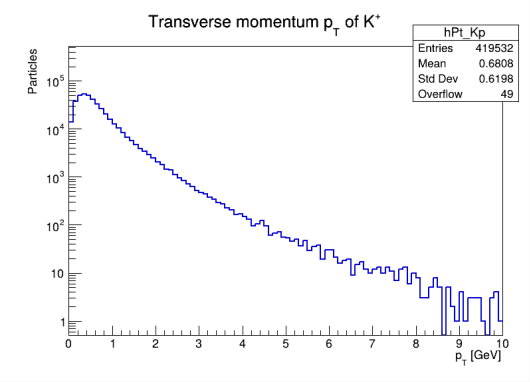


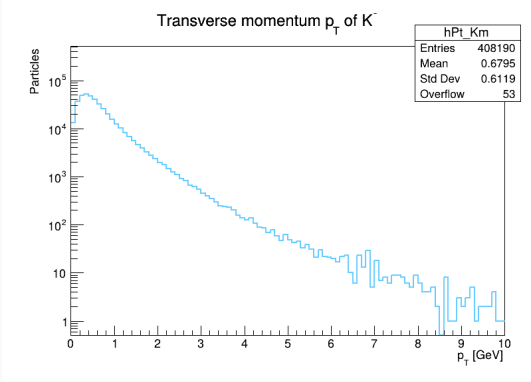
Pęd poprzeczny p_{T}: π^+ i π^-



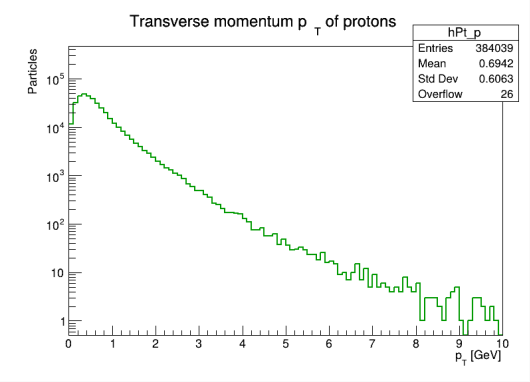


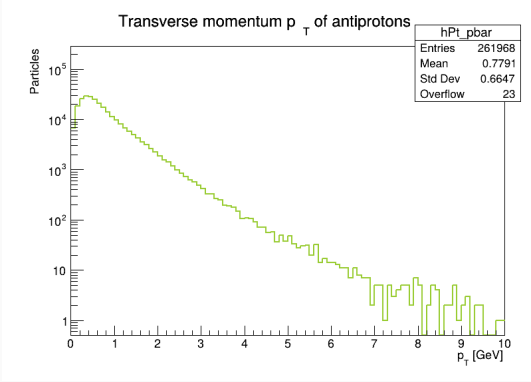
Pęd poprzeczny p_{T}: K^+ i K^-



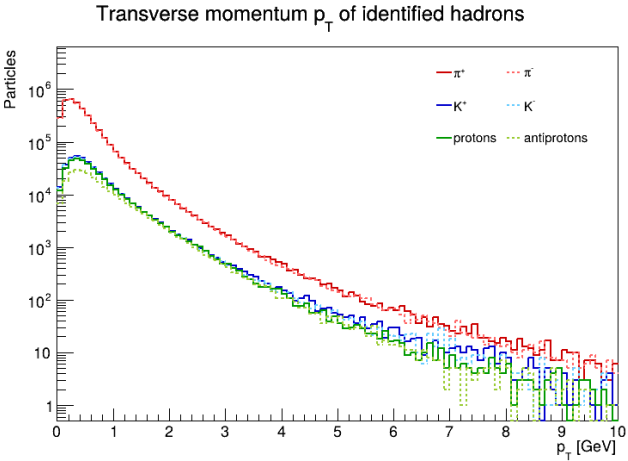


Pęd poprzeczny p_{T}: p i $\bar{p}$

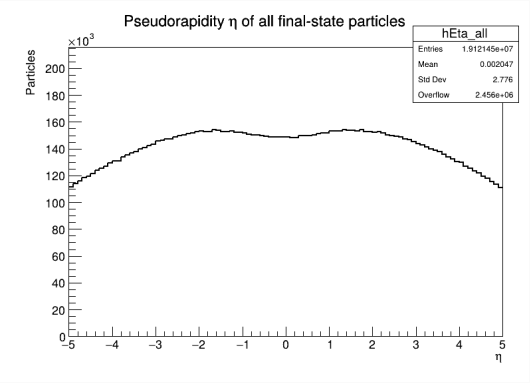


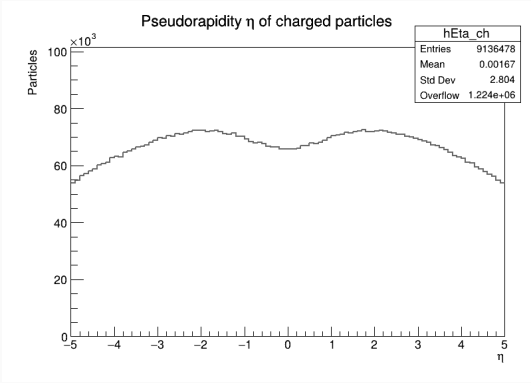


Pęd poprzeczny p_{T}: porównanie

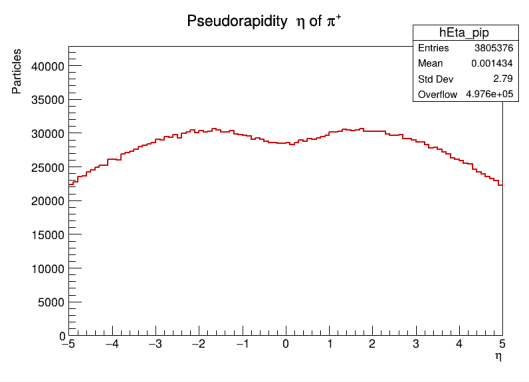


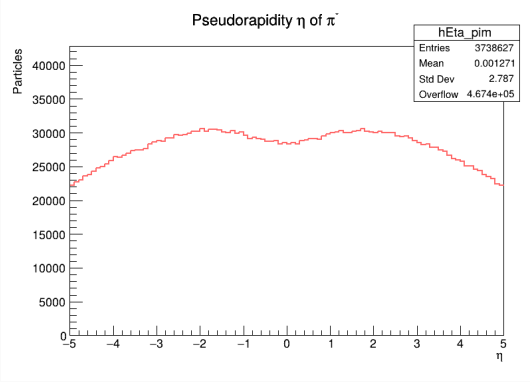
Pseudorapidity η: wszystkie i naładowane



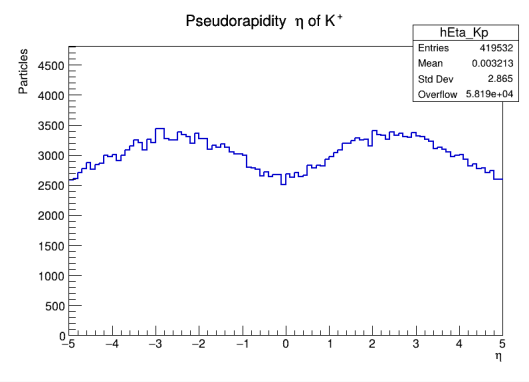


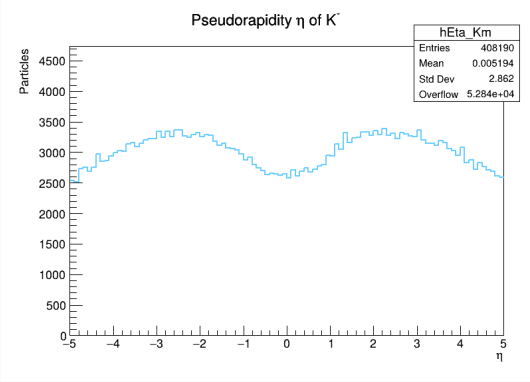
Pseudorapidity η: π^+ i π^-



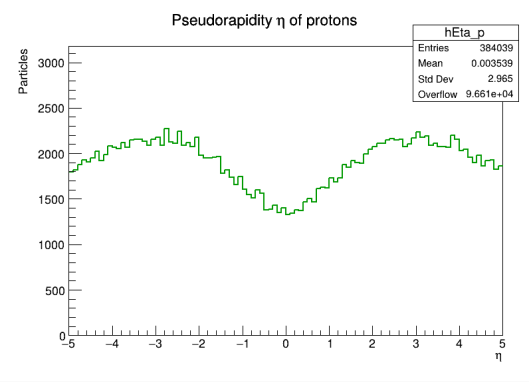


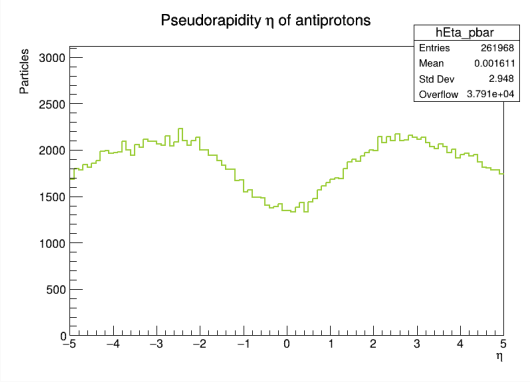
Pseudorapidity η: K^+ i K^-



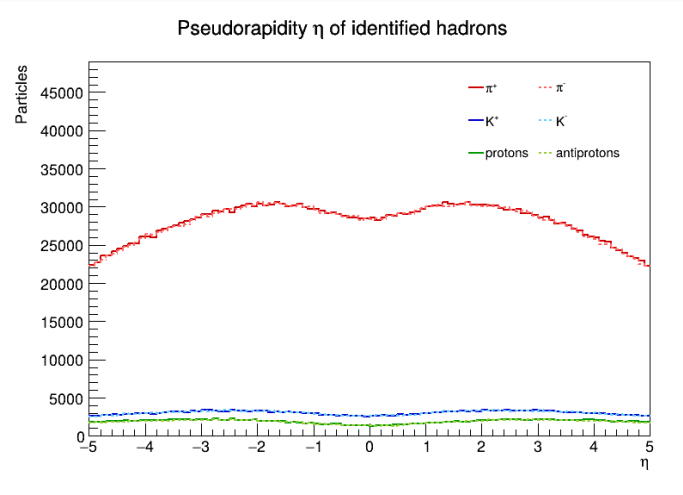


Pseudorapidity η: p i $\bar{p}$

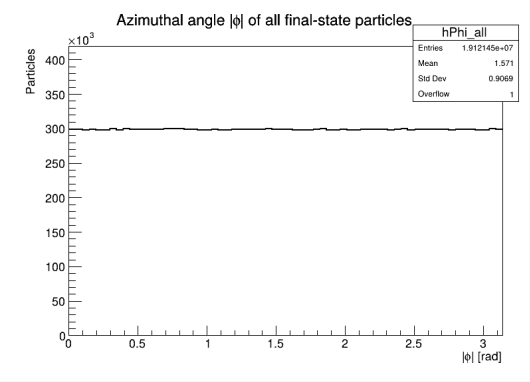


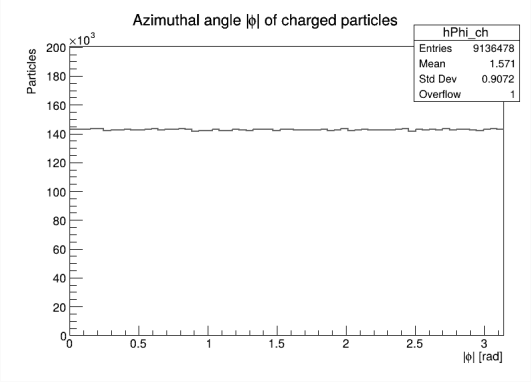


Pseudorapidity η: porównanie

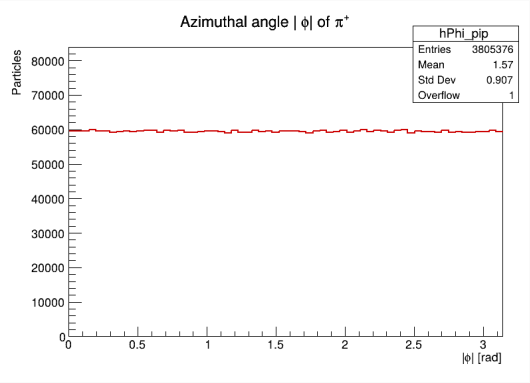


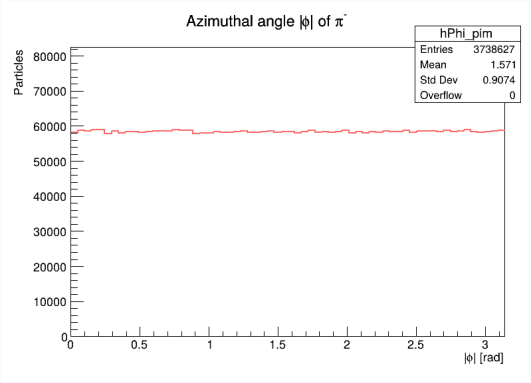
Kąt azymutalny $|\phi|$: wszystkie i naładowane



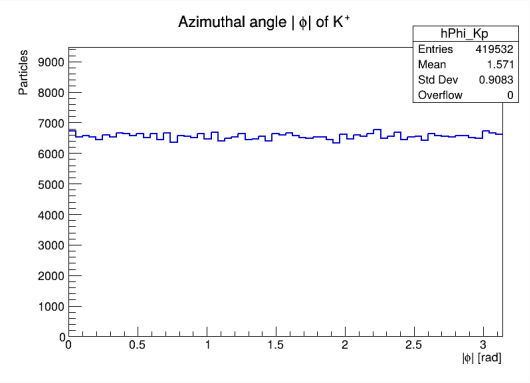


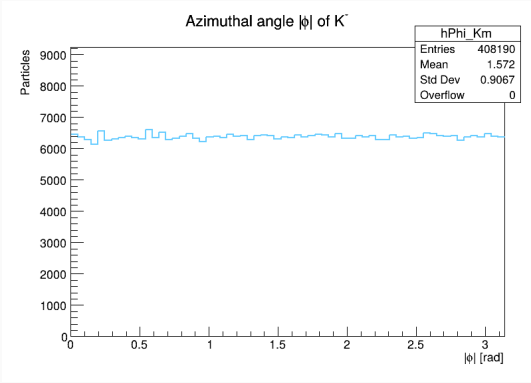
Kąt azymutalny $|\phi|$: π^+ i π^-



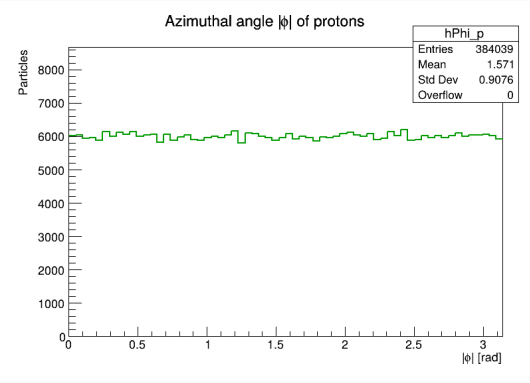


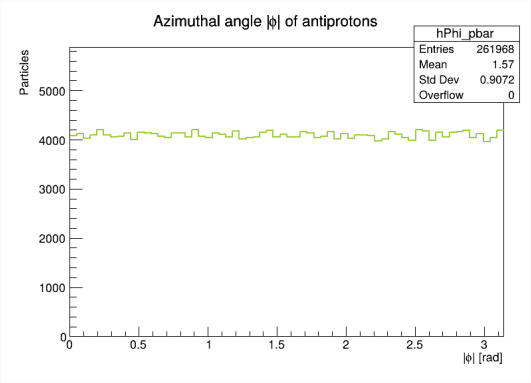
Kąt azymutalny $|\phi|$: K^+ i K^-





Kąt azymutalny $|\phi|$: p i $\bar{p}$





Kąt azymutalny $|\phi|$: porównanie

